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1.

```cpp
#include <iostream>
using namespace std;

int main()
{
    float x, y;

    cout << "Type two different numbers:\n";

    // This checks if the user entered valid numbers (e.g., not letters)
    if (!(cin >> x && cin >> y))
    {
        cout << "\nInvalid input!" << endl;
    }
    else
    {
        // Using the ternary operator to find the maximum:
        // (condition ? value_if_true : value_if_false)
        cout << "\nThe greater value is: " << (x > y ? x : y) << endl;
    }

    return 0;
}
```

2.

```cpp
#include <iostream>
using namespace std;

// Placeholder functions for the actions mentioned in the slide
void action1() { cout << "Executing Action 1" << endl; }
void action2() { cout << "Executing Action 2" << endl; }

int main() {
    char command;
    cout << "Enter command (a or b): ";
    cin >> command;

    switch(command) {
        case 'a':
        case 'A':
            action1(); // Carry out 1st action
            break;
        case 'b':
        case 'B':
            action2(); // Carry out 2nd action
            break;
        default:
            cout << "\a" << flush; // Beep on invalid input
            cout << "Invalid input!" << endl;
    }

    return 0;
}
```

3.

```cpp
int main() {
    switch(command) {
        case 'b':
        case 'B':
            action2(); // Carry out 2nd action
            break;
        default:
            // The '\a' sends an alert sound (beep)
            cout << "\a" << flush; // Beep on invalid input
            break;
    }

    return 0;
}

// Function to display the menu and get input
int menu() {
    char choice;
    cout << "Enter command (a/A or b/B): ";
    cin >> choice;
    return choice;
}

// Action functions
void action1() {
    cout << "Action 1 executed" << endl;
}

void action2() {
    cout << "Action 2 executed" << endl;
}
```

4.

```cpp
void action1();
void action2();

int main() {
    // The function menu() reads a command from the user
    int command = menu();

    // Evaluate the command
    switch(command) {
        case 'a':
        case 'A': // If command is 'a', it "falls through" to 'A'
            action1(); // Carry out 1st action
            break;
        case 'b':
        case 'B':
            action2(); // Carry out 2nd action
            break;
        default:
            // The '\a' character triggers a system alert beep
            cout << "\a" << flush; // Beep on invalid input
    }

    return 0;
}

// Displays the prompt and returns the user's character
int menu() {
    char choice;
    cout << "Enter command (a/A or b/B): ";
    cin >> choice;
    return choice;
}

// First action definition
void action1() {
    cout << "Action 1 executed" << endl;
}

// Second action definition
void action2() {
    cout << "Action 2 executed" << endl;
}
```

5.exercise

```cpp
#include <iostream>
#include <iomanip> // This allows us to use setw() to align the numbers

using namespace std;

int main() {
    cout << "****** MULTIPLICATION TABLE ******" << endl << endl;

    // 1. Print the top header row (numbers 1 to 10)
    cout << "    "; // Space for the side column
    for (int i = 1; i <= 10; i++) {
        cout << setw(4) << i;
    }
    cout << "\n----------------------------------------" << endl;

    // 2. Use nested loops to build the table
    for (int row = 1; row <= 10; row++) {

        // Print the side column number followed by a bar
        cout << setw(2) << row << " |";

        for (int col = 1; col <= 10; col++) {
            // Multiply row by column and align it with a width of 4
            cout << setw(4) << (row * col);
        }

        cout << endl; // Move to the next line after each row
    }

    return 0;
}
```

6.exercise

```cpp
#include <iostream>
#include <cstdlib>

using namespace std;

int main() {
    unsigned int seed;


    cout << "Enter a seed value (0-65535): ";
    cin >> seed;

    srand(seed);

    cout << "20 random numbers between 1 and 100:" << endl;


    for (int i = 0; i < 20; ++i) {

        int randomNumber = (rand() % 100) + 1;
        cout << randomNumber << " ";
    }

    cout << endl;

    return 0;
}
```

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   SPELL CHECKER

39 20 39 38 56 98 66 86 51 13 54 1 43 82 38 22 46 86 98 81

c:\Users\user-46\Desktop\c++\week4>cd "c:\Users\user-46\Desktop\c++\week4\" && g++ 5.cpp -o 5 && "c:\Users\user-46\Desktop\c++\week4\"5
Enter a seed value (0-65535): 20
20 random numbers between 1 and 100:
4 80 74 52 39 96 79 9 3 56 96 82 37 28 29 36 17 74 44 22

c:\Users\user-46\Desktop\c++\week4>
```